Appendix A

Readme File – Report

Introduction:

Welcome to the Hajj and Umrah Analytics with Power BI project.

This README provides an overview of the structure and content of the final project report, which documents the complete process of analyzing Hajj and Umrah data using Power BI. The report presents detailed explanations of the project workflow, dashboard design, data preparation, visual models, implementation process, and analytical findings.

The purpose of this report is to support better understanding of pilgrimage management, crowd analysis, and pilgrim health and behavior through interactive business intelligence techniques and data visualization.

Downloading the Report:

You can access the report by following these simple steps:

1. Download: Retrieve the report document from the provided source (shared folder, email, or project submission platform).
2. **Viewing:** Open the downloaded file using Microsoft Word or any compatible word processing software.

Report Content:

The report is organized into clearly structured sections:

1. Introduction

Provides background information about the importance of Hajj and Umrah management and explains the purpose of developing an analytical dashboard using Power BI.

2. Roles and Contribution of Team Members

Describes the responsibilities and contributions of each project team member during the dashboard development and report preparation process.

3. Problem Description

Explains the challenges related to managing large pilgrim crowds, monitoring crowd conditions, analyzing health situations, and organizing operational information effectively.

4. Overview of Solution

Summarizes how Power BI was used to create interactive dashboards and visual reports that support easier analysis and decision-making.

5. Description of Data

Describes the dataset used in the project, including data related to pilgrims, crowd conditions, transportation, stress levels, health conditions, and activity information.

6. Data Preparation

Explains the data cleaning and transformation processes performed in Power BI, including:

1. Removing irrelevant columns
2. Handling missing values and blanks
3. Changing data types
4. Renaming columns for better readability

7. Visual Model

Provides an overview of the dashboard's visual structure and explains how the visuals were organized to support efficient analysis.

8. Dashboard Design Overview

Describes the dashboard pages, visual elements, navigation buttons, slicers, and design choices used to improve usability and user interaction.

9. Implementation

Explains the implementation process, including:

- Dashboard layout
- KPI cards
- Gauge charts
- Donut charts
- Slicers
- Other visualizations and measures used in the dashboard

10. Power BI Features Used

Lists the Power BI tools and features utilized in the project, such as Power Query Editor, KPI cards, slicers, interactive visuals, navigation buttons, and custom measures.

11. Problems Faced

Discusses technical and analytical challenges encountered during the project development process.

12. Limitations

Highlights the limitations related to the dataset, analysis scope, or dashboard functionality.

13. Future Work

Suggests possible improvements and future enhancements for expanding the dashboard and improving analytical capabilities.

14. Alignment with Sustainable Development Goals

Explains how the project aligns with sustainable development goals by supporting improved crowd management, health awareness, and operational efficiency.

15. Conclusion

Summarizes the project outcomes, analytical insights, and the overall importance of using business intelligence tools in Hajj and Umrah management.9. Implementation: Describes the custom measures and layout strategies applied during the project.
10. Power BI Features Used: Lists the Power BI tools and features utilized to build the dashboard and reports.

Using the Report:
To get the best experience from the report:

5. **Read Thoroughly:** Review all sections to fully understand the project workflow, dashboard design, and analytical findings.
6. **Use the Table of Contents:** Navigate easily between report sections and figures.
7. **Explore the Dashboard Together with the Report:** Using both the report and the Power BI dashboard together provides a deeper understanding of the analysis.
8. **Further Analysis:** The findings can support future research, operational planning, and decision-making related to pilgrimage management.

Conclusion:

We hope this Hajj and Umrah Analytics report provides meaningful insights into pilgrim behavior, crowd conditions, operational management, and health-related analysis.

The report demonstrates how Power BI can be used to transform raw data into interactive and understandable visual insights that support better planning and decision-making.

Feedback and suggestions for improving the project are always welcome.

Notes:

Software Required: Microsoft Word or any compatible document viewer.
Recommended Software for Dashboard Viewing: Power BI Desktop.
Support Contact: Add team contact information or email if needed.